# CYBERGUARD SENTINEL

AI-Based User Behavior Anomaly Detection Platform

REPORT CLASSIFICATION: CONFIDENTIAL
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## 1. Executive Security Summary

This report summarizes behavioral telemetry, machine learning anomaly detection scores, and security rule evaluations across monitored user accounts. Anomalies are detected via Scikit-Learn Isolation Forest paired with behavioral baseline profiling and rule heuristics.

| Total Monitored Events | ML Anomaly Detections | High-Risk Incidents | Open Security Alerts |
|---|---|---|---|
| 47 | 7 | 7 | 9 |

## 2. Risk Score & Distribution Analysis

| Risk Tier | Score Range | Event Count | Proportion (%) | Operational Policy |
|---|---|---|---|---|
| LOW RISK | 0 - 29 | 38 | 80.9% | Standard logging; automatic baseline continuous learning |
| MEDIUM RISK | 30 - 69 | 2 | 4.3% | Elevated monitoring; baseline freeze; SOC analyst triage queue |
| HIGH RISK | 70 - 100 | 7 | 14.9% | Immediate alert raised; session termination / MFA step-up candidate |

## 3. High-Risk & Suspicious Activity Log (Recent 8 Incidents)

| Timestamp (UTC) | User | IP / Location | Target Resource | Risk Score | Flagged Detection Details |
|---|---|---|---|---|---|
| 2026-10-31 21:09 | admin | 185.220.101.5<br>Simulated Lab Node (NEW_IP) | /projects/tickets | **48.3**<br>(Medium) | Impossible Travel Detected: Ingress shifted 6,116 km from Delhi to Frankfurt in 1.6 minutes (Implied Velocity: 229,986 km/h - Commercial Jet Limit: 800 km/h). |
| 2026-10-31 21:09 | admin | 10.0.12.88<br>Simulated Lab Node (MIDNIGHT) | /documents/shared | **87.7**<br>(High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.59).; Login attempt at unusual hour (03:00) deviates from established user schedule. |
| 2026-10-31 21:07 | admin | 127.0.0.1<br>Internal Portal Ingress | /dashboard | **35.4**<br>(Medium) | Login attempt at unusual hour (21:00) deviates from established user schedule. |
| 2026-10-31 19:27 | alice.chen | 45.154.255.89<br>London, United Kingdom | /api/v1/internal/admin-dump | **100.0**<br>(High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.56).; Authentication originated from unrecognized IP address: 45.154.255.89. |
| 2026-10-31 19:27 | alice.chen | 45.154.255.89<br>External Suspicious Ingress | /api/v1/auth/login | **100.0**<br>(High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.77).; Login attempt at unusual hour (04:00) deviates from established user schedule. |

| | | | | | |
|---|---|---|---|---|---|
| 2026-10-31 19:27 | david.kim | 103.152.220.10 External Suspicious Ingress | /financial/payroll | **100.0** (High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.62).; Login attempt at unusual hour (23:00) deviates from established user schedule. |
| 2026-10-31 19:27 | eva.rossi | 91.240.118.23 External Suspicious Ingress | /database/credentials | **100.0** (High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.74).; Login attempt at unusual hour (01:00) deviates from established user schedule. |
| 2026-10-31 19:27 | bob.martinez | 194.26.29.112 External Suspicious Ingress | /admin/system-logs | **100.0** (High) | Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.71).; Login attempt at unusual hour (03:00) deviates from established user schedule. |

## 4. MITRE ATT&CK; Adversarial Tactics & Technique Alignment

Security events, behavioral deviations, and Isolation Forest ML anomalies are correlated directly with official MITRE ATT&CK; Enterprise Matrix v14 tactics and techniques to facilitate standardized SOC response:

| Technique ID | Tactic Category | Adversary Technique Name | Detection Heuristic & Indicator |
|---|---|---|---|
| **T1110.001** | Credential Access | Brute Force: Password Guessing | Adversary repeatedly attempts authentication passwords against valid account names to gain unauthorized access. |
| **T1078.004** | Defense Evasion & Initial Access | Valid Accounts: Cloud / Enterprise Accounts | Adversary leverages stolen credentials from geographically disparate locations, violating Haversine physical velocity limits. |
| **T1083** | Discovery & Credential Access | File and Directory Discovery | Adversary enumerates and probes sensitive application assets including credential stores, system configuration, and payroll vaults. |
| **T1588.002** | Resource Development | Obtain Capabilities: Penetration Testing Tool | Adversary interacts with authentication endpoints via adversarial distributions (Kali Linux, Parrot OS, Tor Browser) or automated scraping utilities. |
| **T1071.001** | Command and Control | Application Layer Protocol: Web Protocols | Adversary establishes authentication session through previously unobserved foreign IP, proxy, or anonymous ASN infrastructure. |
| **T1078** | Defense Evasion | Valid Accounts: Schedule Anomaly | Authentication attempted during anomalous off-hours schedule (e.g. 02:00 AM) to circumvent active administrator vigilance. |
| **T1204** | Execution | User Execution: Rogue Client Device | Session initiated from an unrecognized hardware device signature or spoofed User-Agent environment. |
| **T1078** | Initial Access | Valid Accounts: Multi-Vector Outlier | Isolation Forest identified extreme multi-dimensional divergence across timing, velocity, frequency, and duration dimensions. |

## 5. Recommended Security Actions & Remediation

**1. Multi-Factor Authentication (MFA) Step-Up:** Enforce mandatory FIDO2 / TOTP challenge for logins originating from new IP addresses or unfamiliar client operating systems.

**2. Account Lockout & Throttling:** Configure automated temporary lockout or exponential rate-limiting for accounts accumulating 3+ failed logins in 5 minutes.

**3. Baseline Safeguard Enforcement:** Keep behavioral baseline profiles locked for users undergoing internal investigation to prevent adversarial profile poisoning.

**4. Geofencing & Off-Hours Ingress Policies:** Implement conditional access policies restricting administrative asset access between 23:00 and 06:00 UTC without prior SOC clearance.

**5. SOC Incident Response:** Review all OPEN High and Critical security alerts within a 15-minute SLA. Resolve alerts with thorough forensic notes.